Final Project Design Phase

Computer Science - Week 11

Oct 3, 2026 - Version 0.0.4

For complete instructions and project ideas, refer to the assignment in Canvas. Specifically, review the final project requirements before you continue below. If you have not reviewed the project requirements in Canvas (you might have to go to the modules view and check the current or past week's content), you will not be able to properly complete the parts below. In addition, it may be helpful to review the information on this Drafter documentation page.

0) Project Planning and Preparation

Successful programmers spend time planning before they begin coding. *After* reviewing the project requirements, answer the following questions to help guide your design and development process.

A. What concepts from class or previous assignments might help you complete this project? Which programming concepts (loops, conditionals, lists, classes, etc.) do you expect to use? What programming skills do you want to improve while completing this project?

I expect to use lists, loops, conditionals, classes, and more. I want to improve using outside libraries and general skills that go into making a project.

B. Which parts of the project do you expect to be the easiest for you? The hardest? Why?

I expect writing the drafter code to be the easiest. I expect getting a working map to be the hardest. Also I expect fetching data from the UD bus info website to also be difficult.

C. Even though you have not yet designed the final project, estimate which tasks will likely require the most time and effort? Which features or components should be implemented first?

Making a map will most likely require the most time and effort. I will first need to implement most of the draft in order to get to the point because I will need to be

hosting it to see if it works as expected.

D. What distractions might interfere with your work and how will you minimize them? What resources will you need available while working? What questions, if any, do you still have about the assignment requirements?

Because I am working in VS Code, the suggestions it gives are annoying, so I silence them. Also I tend to do my best work when I don't have people around me so I work in my dorm.

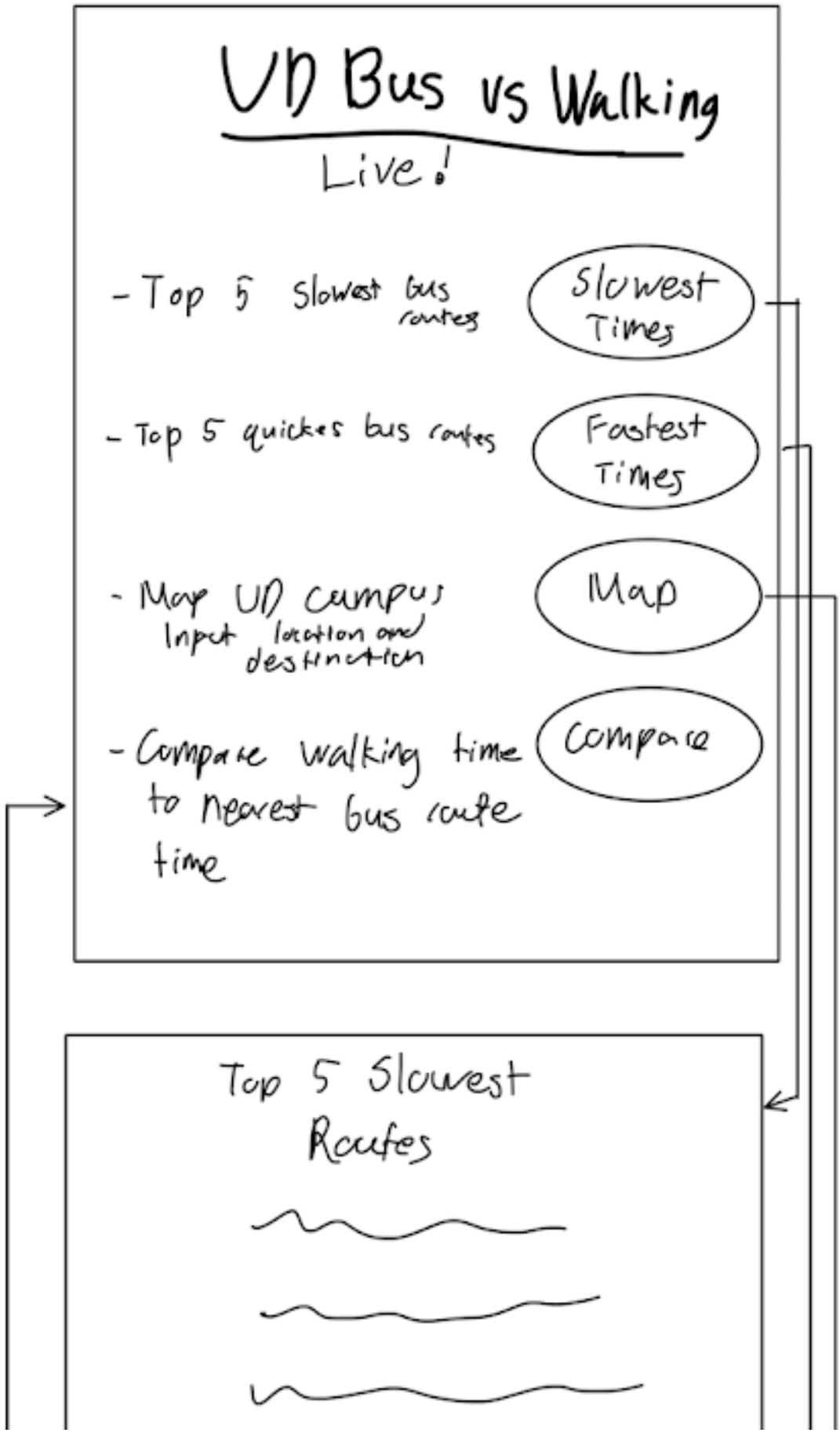
1) Site Diagram

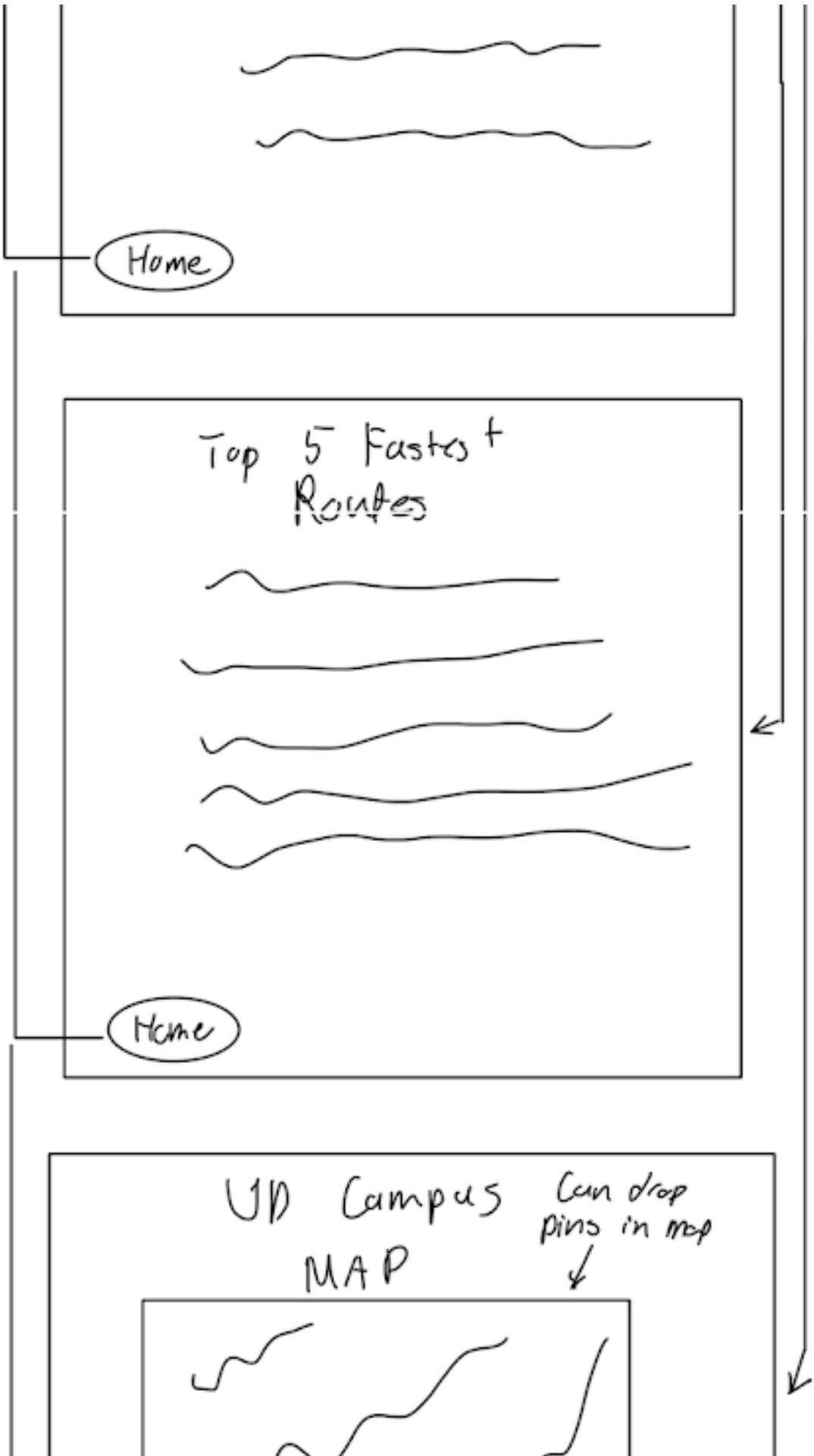
Create a mock-up of your planned website. We suggest you make the mockup on paper, although you can also use software like PowerPoint or Slides to make a mockup. There is other design software that could also be used, but we strongly encourage you to keep it simple. Do not spend more than 15-30 minutes detailing the mock-up. Your goal should be to make a very quick and dirty diagram, to show that you have some kind of idea.

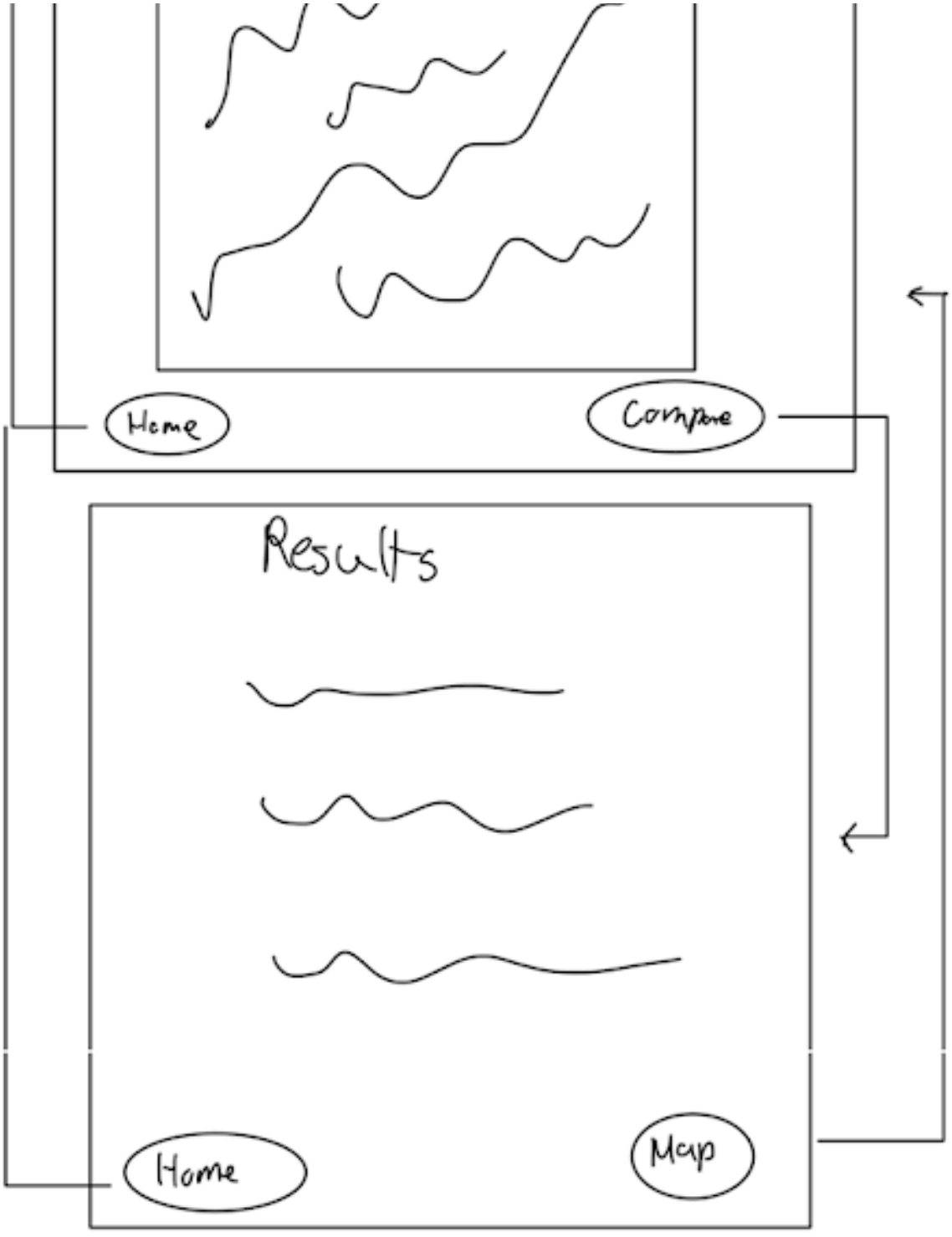
We recommend drawing each page as a rectangle, and laying out the text, buttons, and other components inside. Then, use arrows to indicate links between pages. Label each page the way you will name each function, and then put labels on the arrows to indicate what the button links. Also write down any parameters that should come into the pages.

In addition to the pages and routes, write down what fields your `State` will have, both their names and types. Think clearly about what data will need to be persistent across the entire website.

Add photo(s) or screenshot(s) of your website mock-up and labeled diagram on the next page.


UD Bus vs Walking
Live!
- Top 5 Slowest bus routes
Slowest Times
- Top 5 quickes bus routes
Fastest Times
- Map UD campus Input location and destination
Map
- Compare walking time to nearest bus route time
Compare
Top 5 Slowest Routes


Home
Top 5 Fastest Routes
Home
UD Campus MAP
Can drop pins in map


Home
Compare
Results
Home
Map

2) Implementing Requirements

This table should serve as a draft for how you will implement the requirements into your final project. You may change your ideas later on, but your final site must fulfill all the requirements.

Requirement	**Implementation**
Purpose/Functionality A 1-2 sentence description of the purpose or functionality of your site	**The purpose of this app is to allow users to check whether it is quicker to walk to their destination or to wait for the nearest UD bus and take that instead**
State Dataclass Attributes Minimum of 4 attributes, one of which must be a list/dict of instances of your second dataclass. Briefly describe each attribute, its type, and how it will be meaningfully modified throughout the site.	- Location - This will most likely be a float of coordinates - Destination - This will most likely be a float of coordinates - Walking time - Int, minutes to destination - Bus arrival time - Int, how many minutes away - Route times - List, ints
Second Dataclass Your second dataclass, including its attributes, their types, and how they will be meaningfully modified (if they are modified)	- Is_there_bus - Boolean if there are buses - Nearest - Str, what route is nearest

Routes (minimum 7) The names of your route functions	Show data - Shows the imputed data such as fastest bus routes Show Map - Shows the user the map Update Page - Updates the page with the selected points Go home - Takes the user to the home page Slowest Routes - Take the user to the slowest routes page Fastest Routes - Take the user to the fastest routes page See results - Shows the user the results
Pages (minimum 5) A brief description of the pages in your site	Home Page - Shows the user all of the options of the app Slowest Routes - Shows the user the top 5 slowest bus routes Fastest Routes - Shows the user the top 5 fastest bus routes Map Page - Allows the user to input points on the map Results - Shows the user what is fastesr
User Input (minimum 3) Description of user input components (TextBox, TextArea, CheckBox, or SelectBox) and their purposes For more information on these	- Buttons - One to take the user to each different page - Map - Allows the user input points - Drop downs - There may be drop downs for common

input types, see Drafter Documentation.	routes
If Statements (minimum 3) Brief description of what will be evaluated in your if statements and how they will meaningfully change the logic of the site	- Checks if it can fetch data from the webpage - Checks if buses are running - Checks if the user has valid inputs
Loop (minimum 1) This loop must iterate over the list/dict of dataclass instances in your State attribute. Description of this loop and why it is meaningful to the site	- Will go through all of the bus routes and destinations to see what is closest to the user

LLM Conversation

If you used an LLM in any part of this work, copy and paste every part of that conversation below. Otherwise, you may be given a zero for the assignment. Obviously, these words will not all be yours, and this is the exception to the requirement at the top of this assignment that all responses be in your own words. If you did not use an LLM to assist you in this assignment, simply say so below.

N/A

Final Submission

When you are finished, download this file as a Word Document (docx) and upload the file to the appropriate assignment on Canvas.